

Wrong attack radius calculation damages/kills entities through blocks and corners

MC-2310 Hitboxes

Status	Resolved / Fixed
Affected	Minecraft 1.4.3, Snapshot 13w05b, Minecraft 1.5.1, Minecraft 1.5.2, Minecraft 1.6.1, Minecraft 1.6.2 (+183 more)
Fixed in	23w33a
Reported	2012-11-05

STEPS TO REPRODUCE

1. Build a house from opaque blocks such as planks or cobblestone, with no windows or doors, and keep additional blocks in your inventory.
2. Leave a two-block-tall, one-block-wide empty doorway in one wall.
3. Outside the house, spawn a Ravager.
4. Switch to Survival mode.
5. Wait until the Ravager starts moving toward you.
6. Run into the house through the empty doorway.
7. Wait until the Ravager hits you once through the doorway.
8. Place two blocks to seal the doorway completely.
9. Observe that the Ravager can still damage the player from about two blocks away through the complete wall.

OBSERVED BEHAVIOUR

Actual behavior

After the player enters the house and seals the two-block-high doorway with opaque blocks, the Ravager can continue performing its melee attack from outside. It damages the player repeatedly even though a complete solid wall separates the two entities and there is no unobstructed path between their bounding boxes.

The attack can also occur when the player is positioned around a corner, indicating that the attack-range check is based on the distance between the entities rather than whether the intervening blocks prevent contact. The Ravager therefore continues attacking until the player moves far enough away to leave the calculated attack range, despite remaining physically protected by the wall.

EXPECTED BEHAVIOUR

Mobs should only be able to damage entities when they have a valid, unobstructed attack path to the target. A completely sealed wall or solid blocks around a corner should prevent attacks, regardless of the mob's calculated attack radius or proximity. After the doorway is sealed, the Ravager should stop hitting the player and should not be able to damage or kill them through the blocks.

ENVIRONMENT

Minecraft Java Edition Versions: 1.4.3, 13w05b, 1.5.1, 1.5.2, 1.6.1, 1.6.2, 1.7.2, 1.7.5, 14w11b, 1.7.6-pre1, 1.7.6-pre2, 14w18b, 1.8-pre1, 1.8-pre2, 1.8, 1.8.1, 1.8.3, 1.8.8, 15w33b, 15w33c, 15w47c, 15w49a, 1.8.9, 15w50a, 16w02a, 16w05b, 16w06a, 1.9 Pre-Release 2, 1.9 Pre-Release 3, 1.9 Pre-Release 4, 1.9, 1.9.1 Pre-Release 1, 1.9.1 Pre-Release 3, 1.9.2, 1.9.4, 1.10, 1.10.2, 16w32a, 16w32b, 16w33a, 16w35a, 16w40a, 16w41a, 16w44a, 1.11 Pre-Release 1, 1.11.2, 17w06a, 1.12 Pre-Release 2, 1.12 Pre-Release 6, 1.12, 1.12.1, 1.12.2, 18w03b, 18w07c, 18w21b, 1.13-pre3, 1.13, 1.13.1, 1.13.2, 18w43c, 18w47b, 18w50a, 19w04b, 19w05a, 19w11b, 19w12b, 19w13a, 19w14a, 1.14 Pre-Release 1, 1.14 Pre-Release 2, 1.14 Pre-Release 4, 1.14 Pre-Release 5, 1.14, 1.14.1, 1.14.2 Pre-Release 1, 1.14.2 Pre-Release 2, 1.14.2, 1.14.3, 1.14.4 Pre-Release 1, 1.14.4 Pre-Release 3, 1.14.4 Pre-Release 4, 1.14.4 Pre-Release 5, 1.14.4 Pre-Release 6, 1.14.4, 19w34a, 19w35a, 19w36a, 19w37a, 19w38b, 19w39a, 19w40a, 19w41a, 19w42a, 19w45b, 19w46b, 1.15 Pre-release 1, 1.15 Pre-Release 2, 1.15 Pre-release 3, 1.15 Pre-release 4, 1.15 Pre-release 5, 1.15 Pre-release 6, 1.15 Pre-release 7, 1.15, 1.15.1, 1.15.1 Pre-release 1, 1.15.2 Pre-Release 1, 1.15.2 Pre-release 2, 1.15.2, 20w06a, 20w07a, 20w08a, 20w09a, 20w10a, 20w11a, 20w13a, 20w13b, 20w15a, 20w17a, 20w18a, 20w19a, 20w20a, 20w20b, 20w21a, 1.16 Pre-release 2, 1.16 Pre-release 3,

1.16 Pre-release 6, 1.16 Pre-release 7, 1.16 Pre-release 8, 1.16 Release Candidate 1, 1.16, 1.16.1, 20w28a, 20w29a, 20w30a, 1.16.2 Pre-release 1, 1.16.2 Pre-release 2, 1.16.2 Pre-release 3, 1.16.2 Release Candidate 1, 1.16.2 Release Candidate 2, 1.16.2, 1.16.3 Release Candidate 1, 1.16.3, 1.16.4 Pre-release 1, 1.16.4 Pre-release 2, 1.16.4 Release Candidate 1, 1.16.4, 20w45a, 20w46a, 20w48a, 20w49a, 20w51a, 21w03a, 1.16.5, 21w05a, 21w05b, 21w06a, 21w07a, 21w08b, 21w10a, 21w11a, 21w13a, 21w14a, 21w20a, 1.17 Pre-release 2, 1.17 Release Candidate 1, 1.17, 1.17.1, 21w39a, 21w40a, 1.18 Pre-release 1, 1.18 Pre-release 5, 1.18, 1.18.1 Pre-release 1, 1.18.1, 22w05a, 22w06a, 1.18.2, 22w12a, 22w14a, 22w17a, 1.19, 1.19.1 Release Candidate 1, 1.19.1 Pre-release 5, 1.19.2, 1.19.3, 1.19.4, 23w13a, 1.20.1, 23w32a (includes snapshots)

ORIGINAL DESCRIPTION

The bug

Mobs can attack you through blocks and corners due to miscalculation of attack radius.

How to reproduce

- Build a completely sealed off house with no windows or doors from opaque blocks like planks or cobblestone or the like, leave some blocks in the inventory for you to place in step 7.
- Make a two-block tall one-block wide empty doorway in the wall for you to go through and to seal off with blocks in step 7.
- Outside the house, spawn a Ravager.
- Switch to survival mode.
- The Ravager should start going for you.
- Run in the house, let it hit you once through the empty doorway.
- Seal off the doorway with two blocks.

Result: The Ravager is still able to hit the player, from about two blocks away, with a complete wall every possible place in between.

Expected: The Ravager can't attack the player any more.

Example

An example can be seen in this comment.

Old example

I just created a villager in MCEdit with the Create Shops filter to have a custom shop. I saved it and launched MC. The villager started to run around normally in his house but then he went to the door where a zombie was knocking at (wooden door on normal difficulty) and the villager blinked red and got damage and after a few hits he died. The zombie killed him through a closed door and the villager did not run away he always came back to the door until he died.

Additional informations

From kingsupernova:

There are a number of bug reports about attack radius that are all very similar.

MC-2310

Resolved

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MC-18326

Resolved

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MC-50668

Resolved

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MC-63965

Resolved

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MC-71834

Resolved

, and

MC-74907

Resolved

are all about the attack radius of mobs extending through blocks. (Some mobs are more bugged than others, but it's the same basic problem).

There are also a few related issues:

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MC-1297

Resolved

is the same as the above, but for players.

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MC-3059

Resolved

is the same as the above, but for arrows.

Code analysis

Based on 1.11.2 decompiled using MCP 9.35 rc1

The problem seems to be the method `net.minecraft.entity.ai.EntityAIAttackMelee.ch`

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-2310>